

Using Item Personality-Based Profiling in Music Recommender Systems

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Abstract

Music recommender systems play an essential role in giving music suggestions in accordance to a user's liking. However, there are two common problems with regards to recommendation algorithms, namely user variability and the cold start problem. With this, the researchers aim to address the said problems by incorporating personality profiles to aid in giving music recommendations in accordance to a user's interests. The study made use of the LightFM Model to incorporate the user's top five favorite genres and big five personality scores, hence using a hybrid approach. The hybrid model is then compared with the traditional collaborative filtering model with certain metrics to determine its performance. According to the results, the hybrid model outperforms the traditional CF model, proving that incorporating personality profiles can improve the accuracy of the model in giving music suggestions to a user.

CCS Concepts

• Computing Methodologies; • Machine learning; • Machine learning approaches; • Neural networks;

Keywords

Recommender Systems, Collaborative Filtering, LightFM

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1 Introduction

With the increasing adoption of music subscription services within the community, recommender systems play a crucial role in suggesting songs that align with the user's interests. However, addressing user variability and overcoming the cold-start problem presents significant challenges for traditional recommendation algorithms. Users' music preferences can be diverse and multifaceted, making it difficult for these algorithms to accurately capture their preferences [1].

The researchers will focus on the effectiveness of proportion-based IPPs towards personality-based music recommender systems. The current study by Alharthi focuses on the use of the recommender systems in general, and with this, the researchers intend to apply the said approach to music recommender systems [2]. [3] suggests that item-based recommender systems perform better in terms of rating-based CF recommender systems. Referencing the said study, the researchers intend to explore more on item-based recommender systems.

As the study will revolve around the use of personality-based music recommendation systems, certain gaps are intended to be addressed. Only a few studies have explored the use of personality-based music recommender systems as most research would focus on mood and emotion. Li suggests that personality is one of the factors that determines one's musical preferences, besides mood and emotion [3]. With this, the researchers intend to address the gaps presented by the personality-based recommender system and that is its issues with its accuracy. Researchers tend to address this problem by integrating the system with current collaborative filtering methods. [1] found that having a Big Five personality-based music recommendation system had certain personality biases that made the results irrelevant. With this, the researchers intend to propose a new approach regarding personality-based recommender systems that use the Big Five model. As Alharthi's study produces promising results, the researchers believe that the said study can be incorporated with music recommender systems using the Big Five personality model which could lessen errors posed using the Big Five model.

2 Review of Related Literature

Many recent studies in sentiment analysis gave significant rise to Machine Learning algorithms. While there has been much research on Traditional Machine Learning models specifically on the Online Learning dataset, only a few studies have taken into consideration Deep Learning models. This has been attributed to the exceptional accuracy of traditional models.

Lu & Tintarev's study focuses on addressing the cold-start problem amongst recommender systems. With this, their paper focuses on giving better recommendations by re-ranking the suggestions by the recommender and improving the accuracy of the model by integration of personality traits. Their results suggested that the said focus points of the paper have been met, wherein personalized recommendation lists are higher in diversity and lead to an increase in user satisfaction [4].

Klec et al's studied how personality traits can have an influence on music recommendation error. The study collected responses from 279 participants to gather their listening traits and their personality profiles. With this data, the combination between the personality traits and the audio features from the songs were studied to arrive at a conclusion with regards to music recommendation error. Findings suggested that some personality facets had yielded the lowest recommendation error for BFI-2, compared to the higher order of the big five personality traits. The researchers had concluded that this discovery can help other researchers to filter the personality questionnaire down to only the essential questions to be used in gathering the personality profiles of the users to predict their listening habits [1].

Garg & et. al. [5] focuses on developing a music mood-based recommendation system using machine learning and deep learning techniques, with a particular emphasis on addressing the common struggle people face in finding music that matches their mood or emotions. The research utilizes the FERPlus dataset and Spotipy, a Python library for Spotify, for web scraping and data manipulation. The Valence-Arousal theory is employed to classify the song database into mood quadrants, and K-Means clustering is used for further categorization. Features such as valence, energy, danceability, loudness, tempo, speechiness, instrumentality, and popularity are considered. A Deep Convolutional Neural Network (CNN) is employed for classification, with varying accuracies reported for different models and emotions. The model aims to provide real time playlist recommendations based on mood, contributing to mood enhancement through AI-driven song selection. The study suggests that this method surpasses genre classification in accuracy, with potential applications in improving social networking and customer satisfaction. The reported accuracy percentages include below 65% for a Simple CNN with six emotions, above 70% for a Simple CNN with four emotions, below 72% for a Deep CNN with six emotions, and about 80% for a Deep CNN with four emotions. Loss percentages, based on 60% training data and 50% validation data, decrease to 30% or below as the number of epochs increases for both training and validation data.

Joy & et. al. [6] is rooted in the Big Five Model, leverages data obtained from Twitter, encompassing 395,056 listening events, 18,310 users with personality values, and 15,753 unique tracks. The research investigates how users' personality traits influence the

accuracy scores of state-of-the-art recommendation algorithms and explores user bias related to personality traits. The methodology involves data cleaning, personality estimation, and data segmentation for track and artist information extraction. Sparse Linear Methods, Embarrassingly Shallow Autoencoders (EASE), and Variational Autoencoders (Mult-VAE) are utilized for analysis. The reported recall values at different positions (5, 10, and 15) provide insights into algorithm performance based on users' personality traits. For instance, at Recall@5, EASE achieved 0.0366, SLIM achieved 0.0334, and Mult-VAE achieved 0.0433. These rates indicate the proportion of relevant items retrieved by the algorithms at the specified positions. The study recognizes its limitations, such as generalizability and data completeness issues, and outlines future research directions to better understand the causes of biases and explore additional aspects of recommendation quality beyond accuracy.

Delando & et. al. [7] is centered around the MBTI, utilizing the MARSYAS Dataset along with the Extended and S Datasets, the study aims to predict personality traits, specifically extraversion and introversion, based on individuals' music preferences. The methodology involves data collection, dimensionality reduction, normalization, and segmentation based on "Personality Type." Audio frames are segmented, and features are extracted using MIRtoolbox and a variational autoencoder. Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), and Naive Bayes classifiers are employed for analysis. The SVM classifier achieves a performance rate of 71%, showcasing its effectiveness in predicting personality traits based on music preferences. Furthermore, the study explores the correlation between music genres and personality types, revealing a scoring distribution of 53% for extraverts and 47% for introverts, shedding light on the nuanced relationship between music preferences and personality traits.

Dorochowicz [8] is focused on exploring emotions and physiological signals, utilized the DEAM and PMEmo datasets to investigate the intricate relationship between music and human emotions. The research aimed to quantify this connection and develop a machine learning model capable of predicting both music mood based on audio signals and human emotion based on physiological signals. The methodology involved downscaling, format conversion, regularization, and shrinkage, with feature extraction encompassing zero-crossing rate, energy, entropy of energy, and various spectral features. Machine learning models, such as SVM, Random Forest, Decision Tree, Extra Tree, k-NN, MLP, AdaBoost, ElasticNet, Ridge, Lasso, and Adaptive Boost, were employed for analysis. ANOVA results indicated significant variance between people and items, emphasizing individualized emotional responses to music. The Random Forest model demonstrated notable performance in terms of action call count and training time. The arousal and valence analysis revealed distinct emotional responses associated with different music moods. The research achieved an accuracy rate of 81%, underscoring the practical potential of the developed model, including personalized playlist generation. However, the study acknowledged the need for further research and refinement to enhance accuracy and broaden the scope of emotion recognition.

Liu & et. al. [9] utilizes a dataset from the myPersonality website, aimed to develop an algorithm to determine individuals' music tastes based on their personality traits. The approach involved converting Facebook status updates into vector representation. Various

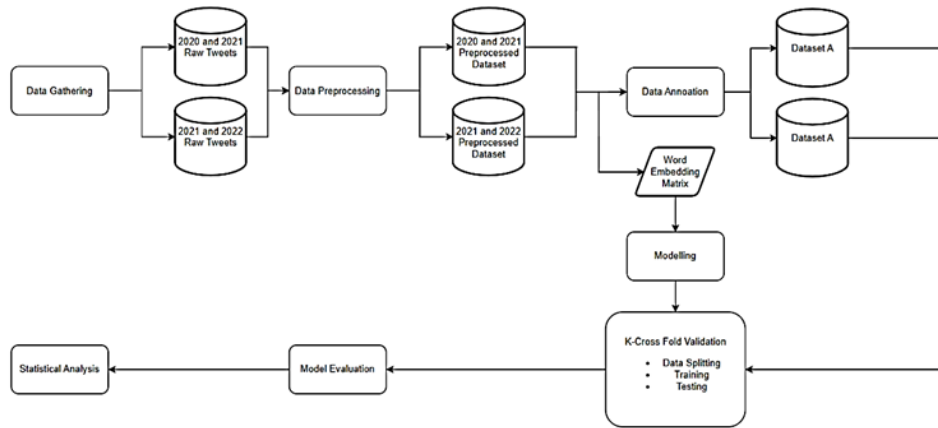


Figure 1: Proposed Model Workflow

recommendation models, including Naive Bayes Classifier and collaborative filtering methods, were employed. The recommendation model's performance was assessed using Root Mean Squared Error (RMSE), with Matrix Factorization achieving the lowest RMSE at 0.88. F-Measures for Big Five Personality Classes, including Openness (0.2308), Conscientiousness (0.4865), Extraversion (0.6349), Agreeableness (0.5000), and Neuroticism (0.6154), were also calculated. The recommendation model outperformed user-to-user collaborative filtering with a rating matrix, although matrix factorization remained superior. The study provided insights into the relationship between personality traits and music preferences, emphasizing the potential of personalized recommendation systems.

Lu & et. al. [10] utilizes the myPersonality dataset and the PsychoFlickr dataset, focused on developing an emotional recommender system that predicts music taste based on personality traits. The user recognition problems were categorized into five binary classifications and further divided into classes (low, medium, high). Forward Features Selection with 10-fold cross-validation was employed, and various classifiers, including Naive Bayes, Logistic Regression, SVM, Random Forest, and GBT with deep networks, were used. The accuracy results for the myPersonality dataset using the proposed approach were O - 0.71, C - 0.72, E - 0.75, A - 0.73, N - 0.71. The self-assessed accuracy results for the myPersonality dataset were lower (O - 0.54, C - 0.55, E - 0.54, A - 0.54, N - 0.54). For the PsychoFlickr dataset, the proposed approach achieved accuracies of O - 0.61, C - 0.59, E - 0.56, A - 0.60, N - 0.61. The self-assessed accuracy results were again O - 0.54, C - 0.55, E - 0.54, A - 0.54, N - 0.54. The experiments demonstrated the success of using user personality for mood recognition in predicting music taste, urging further exploration in this field.

3 Methodology

As shown in figure 1 the procedure in developing and evaluating the model.

3.1 Data Collection

The data collection process will be conducted in the form of a survey. The survey will be used for the explicit gathering of the information required for building the three matrices (user, item, interaction) that will be used for the LightFM Hybrid Model. The questionnaire will have a combined total of 111 questions consisting of the 10-item BFI Questionnaire (Likert Scale of 1 – 5), 100-item Song Rating (1 as like, 0 as dislike), and Preferred Genre Checklist (10 genre options, choosing a maximum of 5).

3.2 DATA PRE-PROCESSING

3.2.1 User Personality Profile. The respondents' answers to the Big Five Inventory (BFI) will then undergo a multi-step process with only the resulting T-score being stored in the dataset. First, raw scores are calculated for each of the five core dimensions (Extraversion, Neuroticism, Agreeableness, Conscientiousness, and Openness) by summing an individual's responses to the relevant questionnaire items. These raw scores are then transformed into the variable "user BF dimension mean" which is used to control for potential discrepancies in question difficulty and establishes a standardized baseline for comparison. This is computed by dividing the item scores by the number of questions associated with each dimension. Following the calculation of user BF dimension means, the BFI employs standardization procedures to facilitate meaningful comparisons across individuals and populations. Z-scores are derived to represent the deviation of an individual's score from the average population score for a specific trait, expressed in terms of standard deviations. Finally, for enhanced interpretability, Z-scores are converted into T-scores with a mean of 50 and a standard deviation of 10. This will result in 5 T-scores, which will be converted into degrees to categorically represent the user's personality profile. A BF score is stored as "high" if it is greater than 55, "low" if it is lower than 45, or average otherwise as demonstrated in Table 1. This will be represented in the appropriate format shown in Table 2 to make it compatible as a user-feature embedding.

3.2.2 Item Features. The researchers will incorporate the proportion-based IPP method used in [2] where the item personality profile is represented in vector form, consisting of 15 features. Each dimension is represented by three features that correspond to degrees (low, average, and high). The value of one feature is the proportion of users who like an item in the degree of one BF dimension which is calculated based on the following formula:

$\text{Proportion} = (\text{total Number of Users who liked the Item and has the same BF degree}) / (\text{total Number of Users who Liked the Item})$.

Spotify API will be used to extract content-based features to be used alongside the item personality profile as item feature weights. Instead of the musical and the standard features (bpm, tempo, release date, volume) that are commonly covered in existing studies, the study will focus on extracting the following sonical components or features that quantitatively describe aspects that contribute to the overall sound and mood of the song, which are Acousticness, Danceability, Energy, Instrumentalness, Liveness, and Valence.

3.2.3 Interaction Matrix. The interactions matrix will store recorded interactions for all users. This will comprise three columns, two of which are foreign keys that link to both the user features matrix (User ID) and song features matrix (song ID). The isLiked column is used to represent interactions as either 1 or 0. To account for the differences between the way interactions are stored in the survey dataset and the Million Song Dataset (MDS), an additional column Weight will be used as an additional feature for model training, assigning different levels of importance depending on the value of a specific interaction. All interactions from the Mapuan students will automatically be assigned the highest value (5) since these represent explicitly liked songs from the provided song list.

MDS user interactions on the other hand are in the form of play counts instead of likes which require a different approach. The researchers employed a min-max scaling transform to normalize the data between a range of 0-1. The scaled values are then multiplied by 4 and rounded to the nearest integer to map to 5 weight levels. This will allow the model to prioritize learning the correlated song features in songs with higher play counts when used as weights during model training.

3.3 Model Development

A hybrid recommender system will be developed utilizing the LightFM library which integrates elements of both collaborative filtering and content-based filtering. In this model, users and songs are represented by latent factors and biases, derived from user attributes, item attributes, and historical interactions. Specifically, users are described through features such as genre preferences and user personality profiles (Big Five) while songs are characterized by their sonic features and the computed Item Personality Profiles (IPPs). With the described preprocessing and feature extraction steps, personality features are more deeply integrated into both the user and item feature matrix, particularly in situations with limited explicit user-item interactions. This will be combined with the subs

For all models, researchers will use LightFM's cross-validation feature to randomly split the interaction set into two disjoint training and test sets with an 80-20 split. The model will then be fitted for 30 epochs and a 0.02 learning rate. The researchers will also test

which of the two loss functions (BPR and WARP) will perform better on offline evaluation before finalizing the model to be evaluated by new users. These two loss functions were chosen for evaluation since both are particularly useful in cases where only positive interactions (likes) are recorded but with different optimizations. WARP is suited for optimizing the top recommendation list, while BPR is more suited for optimizing the AUC score. For model evaluation, the k value is set to 10 to evaluate its performance when generating the top 10 recommendations.

3.4 Model Validation and Evaluation

The researchers will use the built-in evaluation metrics provided by the LightFM library. These metrics can be collectively used to assess the model's performance by computing these for the training and test split of the interaction's matrix. The metrics to be used will be the ROC AUC Score, Precision, Recall, and F1-score. ROC AUC Score quantifies the probability that a randomly selected positive example will have a higher score than a randomly chosen negative example. It measures the model's ability to rank positive examples higher than negative examples. A perfect score is 1.0, indicating optimal performance in distinguishing between positive and negative examples. Precision at k measures the fraction of relevant items among the top k items recommended by the system. It evaluates the precision of the recommendations by considering how many of the top recommendations are relevant to the user. A perfect score is 1.0, indicating that all top k recommendations are relevant to the user's preferences. Recall at k evaluates the system's ability to retrieve all relevant items among the top k recommendations. It measures the proportion of relevant items that are successfully retrieved by the system compared to all relevant items in the dataset. A perfect score is 1.0, indicating that all relevant items are retrieved among the top k recommendations.

The proposed hybrid model as shown in Procedure C will also be evaluated against a pure collaborative filtering (CF) model (based only on interactions without incorporating personality IPP and UPP features) that is trained on the same combined dataset and parameters.

3.5 Online Evaluation

In addition to these metrics, researchers will also employ online evaluation methods to conduct model comparisons from the perspective of new users whose personality and preference counts weren't included in model training. This will be conducted in the form of a web app, which will require 20 new users to take the personality test and input their top 5 genre preferences. This will be used to test the efficacy of generating real-time recommendations before acquiring a list of their liked/disliked songs. The users will then be presented with the generated top 10 recommendations from the two trained models which they will evaluate in the form of a post-feedback questionnaire which is adapted from a study evaluating their personality-based recommender system (Lu et.al, 2018). Moreover, users will also be able to individually rate each song in the generated list as like/dislike. These measures will be precision and recall (like the dataset evaluation), and user feedback. For user feedback, the system is evaluated qualitatively by prompting users to express their opinions on all lists based on the three main

Table 1: Model Metrics Comparison (Train)

Metrics	Collaborative Filtering(CF)	Hybrid BPR	Hybrid WARP
Precision	0.24	0.16	0.23
Recall	0.86	0.02	0.75
AUC Score	1.00	0.67	0.82
F1 Score	0.38	0.04	0.35

Table 2: Model Metrics Comparison (Validation)

Metrics	Collaborative Filtering(CF)	Hybrid BPR	Hybrid WARP
Precision	0.48	0.42	0.46
Recall	0.74	0.22	0.64
AUC Score	0.56	0.53	0.57
F1 Score	0.58	0.29	0.54

aspects based on Pu, Chen, and Hu’s (2011) ResQue User-Centric Evaluation Framework. These questions are in the form of a 5-point Likert scale (1 being strongly disagree and 5 being strongly agree), with one question each for both models being evaluated which will assess Recommendation Quality, Recommendation Diversity and User Satisfaction. Respondents will also be asked which model is more suited for them in terms of recommendation quality and diversity.

4 Result and Discussion

The researchers collected data from 208 students and used said data to feed information into the model, generating recommendations tailored according to their personality scores and genre preferences for model A (hybrid model), and popularity-based recommendations for model B (pure CF model). The precision, recall, AUC, and F1 scores are then computed to evaluate the metrics of the models, for both offline and online evaluation methods. A two-tailed t-test and chi-squared test were also conducted for the survey results from the online evaluation method to verify how statistically significant the results are, with $p < 0.05$.

4.1 Offline Evaluation

As shown in table 1 and table 2, pure collaborative filtering model has the overall best performance metrics when it comes to the training set particularly for recall and AUC. The higher discrepancy between recall and precision score for this model (also referred to as Model B) is indicative of the limitations of this kind of recommender since it prioritizes breadth/diversity of recommendations as opposed to actual relevance(precision) with an emphasis on prioritizing songs with more interactions. The drastic drop in performance when comparing the training and validation/test set for this model also indicates that model’s design has issues with overfitting and is less capable of adapting to unseen data compared to both variants of the hybrid model. Overall, the hybrid models are observed to have good metrics with no large discrepancy between validation and training set, which indicates that it can identify patterns and tailor recommendations accordingly even on data with no recorded interactions. Between the two loss functions tested by

researchers, the hybrid model that uses WARP has the best performance on all metrics. The improved performance of WARP over BPR may be due to the limitation of pairwise loss function when used on relatively sparse and imbalanced data, with the dataset having complete user personality information for less than 20% of the overall userbase. Moreover, it is especially suitable for improving a recommender’s ability to rank items, which is why it was chosen as the final model to represent the Hybrid Personality IPP Model (Model A)

4.2 Online Evaluation

The online evaluation method involves the use of a survey that takes the user features of a user as inputs and then through those inputs, the model generates a list of song recommendations. With this, users are tasked to compare the song recommendations for model A(Hybrid) and model B (Pure CF) by marking the songs in the list that they like. Then, they are tasked to answer a second survey related to the recommendation quality (Q1), diversity (Q2), and user satisfaction (Q3) of the two lists, and compare which list has better recommendations and/or is more diverse. Respondents were also asked to like songs that were recommended to them by the two models. After this, another survey was conducted to determine the quality and diversity of the recommendations of the two models. In response to this, a t-test was conducted to determine the significant difference between each question. The result is shown in Table 3.

Results show that Model A performs better than Model B in terms of Recommendation Quality and User Satisfaction. In contrast to this, Model B outperforms Model A in terms of Recommendation diversity. Looking at the results, users think that model A has better recommendation quality and is more satisfied with its recommendations, whilst the users would prefer model B in terms of diversity. Together with this, a chi-test is conducted for the last two questions, wherein the respondents are asked which of the two lists have better recommendation quality, and which list is more diverse. Shown below is the table for Observed and Expected values. Observed values are the data collected from the respondents whilst Expected values are the calculated values. The result is shown in table 5.

Table 3: Model Metrics Comparison (Train)

	Recommendation Quality		Recommendation Diversity		User Satisfaction	
	AQ1	BQ1	AQ2	BQ2	AQ3	BQ3
Mean	3.7	3.4	3.75	3.95	4.15	3.55
P(T<=t) two-tail	0.428290293				0.518979715	

Table 4: Chi-Test Result (Observed)

	A	B	Hard to Tell	TOTAL
LCQ1	14	4	2	20
LCQ2	3	12	5	20
TOTAL	17	16	7	40

4.3 Precision and Recall

Both models initially produced recommendations with no variation due to the limited song dataset, this also has the benefit of making it easier to identify how the incorporation of personality and other genre features influence the quality and diversity of the recommendations. Pure collaborative systems such as Model B, which solely relies on interaction counts, generated the same set of recommendations based on its overall popularity among the users included in the dataset. The recommendation list is therefore shown to favor older tracks due to inherent popularity bias and does not adjust based on the provided information such as genre and sonic features. In contrast, Model A has personality-based IPPs alongside sonic metadata and genre preferences, which enables it to tailor recommendations even with the limited dataset. Despite the number of users increasing (Dataset A vs Dataset B), Model A was still not able to create more variations. This may be due to the limited number of songs that the user must choose from (100 songs) in the creation of the dataset. For instance, newer songs like Vampire (absent from the 2011 MSD dataset due to zero prior interactions) appeared in the Top 10 by leveraging feature similarities to other tracks already popular among Mapuan students. This has an unintended effect of making it appear less “diverse” to users since the recommended songs are likely to be more specific and like each other as which explains the discrepancy between opinion and metrics-based diversity.

Based on the individual songs feedback for the models’ recommendations, the computed precision and recall values also indicate that Model A outperforms Model B in terms of recommendation quality and diversity which is even more apparent when there is more information available in the dataset. Model A has an average precision value that is 0.13 higher than Model B, which indicates that even with no variation, it has greater accuracy in adjusting the ranking to prioritize items users liked based on the provided IPP features. While in the survey itself, the new respondents seem to perceive Model B as more diverse and have more quality, the computed average recall metrics suggest that Model A also performs better in generating lists that represent a wider coverage to represent variations in user preferences. As a result, Model A is also consistently shown to have higher F1 scores, especially with an increase in dataset size, which is a good indicator that the perceived

recommendation quality is consistent with the actual quality of the recommender which is based on the user’s explicit likes.

5 Conclusion and Recommendation

Overall, the researchers accomplished the primary objective of this study which is to develop a personality-based IPP recommender system as well as effectively demonstrate its potential compared to traditionally pure collaborative filtering recommenders even with the limitations of dataset skewness and sparsity. In both offline and online evaluation, hybrid models outperform pure CF models in both computed metrics (Precision, Recall, AUC, and F1 Score) as well as user-prompted feedback (for Recommendation Quality and Diversity). The output recommendations also suggest that the hybrid model achieved a better balance between relevance and diversity by leveraging personality alongside genre and sonic features to find emergent patterns when combining user, song, and interaction features. To explore the idea more effectively, the researchers recommend acquiring even more samples for the model to learn from and fix the issues that arise from overfitting, specifically more songs for the user to evaluate in creating the dataset and increasing the number of respondents for the dataset. Feature engineering could also be amended to achieve a more reasonable balance between popular and diverse items with the incorporation of personality-based IPPs. Future research can also explore alternative methods for incorporating personality data, such as using other recommender libraries such as scikit instead of LightFM which would have a different method for handling the user-item information matrix and could potentially lead to better performance.

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Table 5: Matrix Comparison

	Precision	Recall	F1 Score
Model A	0.57	0.62	0.58
Model B	0.41	0.46	0.40

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